

E-Rakshak 2026 — PS-4

Weekly Progress Documentation

Team HackersAPK | SentinelScan | Neil Banerjee

Week 1 & Week 2 Report
08 July 2026 — 16 July 2026

WEEK 1 — Problem Understanding & Planning

Wed 08 Jul — Thu 09 Jul 2026 (2-day week, foundation sprint before first mentor meeting)

Objective

Lock project scope, stand up the repository and infrastructure foundations, and finalize architecture before the team's Thursday mentor check-in.

Deliverables Completed

✔ Repository Structure

GitHub repository created at `neil-data/Team-HackersAPK` , structured by architecture layer so each team member's module has a dedicated home:

```
ps4-malware-suite/
├── ingestion/
├── static-analysis/
│   └── yara_rules/india_scam_rules/
├── dynamic-sandbox/
├── agents/
│   ├── orchestrator/
│   ├── mitre_mapper/
│   ├── capability_classifier/
│   └── narrative_agent/
├── storage/
├── backend/app/
├── frontend/src/
├── docs/
└── infra/
```

`.gitignore` , `docker-compose.yml` (Postgres + Redis + Elasticsearch stubs), `.env.example` , and `README.md` all populated. — Neil

✔ Scope Locked — 3 Core Differentiators

- 1. **India-specific scam detection rules** — YARA/behavioral signatures for light-bill, loan-app, and e-Challan/RTO fraud patterns explicitly named in the PS
- 2. **LangGraph agent correlation + plain-language narrative agent** — turns raw static+dynamic+network signals into an officer-readable report
- 3. **Hash-chained, signed chain-of-custody reporting** — evidence-grade, court-ready output, its own visible dashboard module

Stretch goals (Week 4 only, not committed scope): 24/7 live GCP sandbox exposure, encrypted-traffic metadata analysis. — All (team decision)

✔ Architecture Finalized

Six-layer pipeline diagrammed in draw.io (`infra/PS4_Architecture.drawio`): Ingestion & Isolation → Static Analysis → Dynamic/Sandbox Analysis → Behavioral Correlation & Agentic Reasoning → Storage → Output. Diagram iterated once to fix auto-routed edges crossing box labels; rebuilt with clean layer-to-layer container connectors. — Neil

✔ Team Roles Assigned

Member	Ownership
Member 1	Static analysis & ML — APK/PE static engine, YARA rules, IOC extraction
Member 2	Dynamic sandbox & cloud infra — GCP, KVM/CAPE, Android-x86, network containment
Neil (Member 3)	Agent layer — LangGraph orchestrator, MITRE mapper, capability classifier, narrative agent
Member 4	Dashboard, reporting & storage — React frontend, DB schemas, IOC export

✔ Cloud Infrastructure — In Progress

Deployment model decided: two-plane architecture (public Control Plane + isolated Detonation Plane). Google Cloud account creation and `n2-standard-4` instance provisioning (nested virtualization, Mumbai region) assigned to Member 2, using GCP's \$300/90-day free trial credit — projected cost for full hackathon timeline: \$0. — **Member 2 (in progress)**

✔ Sample Sources Identified

Free/legitimate sources for test malware samples confirmed: MalwareBazaar (abuse.ch), theZoo, Koodous. Safe-handling protocol set — samples only ever downloaded onto the isolated GCP sandbox, never the local host machine; `dynamic-sandbox/samples/` path is gitignored. — **Neil**

✔ Frontend Stack Decision

Decided to hand-code the frontend (landing page, login, dashboard) rather than use AI page-builders, for full control over the report-view UX. Auth approach locked: JWT + FastAPI (custom-built), not a third-party provider like Firebase.

Week 1 — Not Yet Started (carried to Week 2 kickoff)

- Docker skeleton per-service `Dockerfile`s (deferred until static-analysis and sandbox owners are actively coding)

WEEK 2 — Core Build Sprint

Fri 10 Jul — Thu 16 Jul 2026

Objective

Build the static analysis engine, ingestion/isolation layer, GCP sandbox infrastructure, and stand up the agent orchestrator skeleton against mock data — target milestone: static pipeline demo + CAPE detonating live on the GCP sandbox by Thursday's mentor meeting.

Deliverables Completed

✔ LangGraph Agent Orchestrator — Working Skeleton

— Neil

Built and tested end-to-end in `agents/orchestrator/` :

```
agents/orchestrator/
├─ schema.py           (Pydantic data contracts)
├─ orchestrator.py     (LangGraph graph definition)
├─ requirements.txt
├─ mock_data/
│   └─ static_analysis_sample.json
```

Graph flow: `load_static_analysis` → `mitre_mapper` (stub) → `capability_classifier` (stub) → `compute_risk_score` → `narrative_agent` (stub) → END

Verified test run against a mock loan-app-scam sample:

Output	Result
MITRE techniques mapped	T1517 (Access Notifications), T1071 (Application Layer C2)
Capability tags	sms_otp_theft, gps_tracking
Risk score	80 / 100
Narrative summary	Stub template generated successfully, real LLM logic slated for Week 4

All stub logic clearly marked with real-implementation notes for Week 4 (MITRE mapping, capability classification, and narrative generation currently use simplified rule-based logic in place of the full agent reasoning and LLM calls).

Committed and pushed to `main` . One hiccup resolved: `mock_data/static_analysis_sample.json` was initially missed on first commit (recreated and verified via `git status` before second push).

✔ Static Analysis Output Schema — Proposed

— Neil, pending confirmation from Member 1

Full schema for static-analysis module output drafted in `schema.py` (`StaticAnalysisOutput`) and demonstrated via the mock JSON sample — covers hash reputation, YARA matches, Android manifest/permissions, PE analysis fields, extracted strings/IOCs, and ML classifier output. Proposed to Member 1 as the integration contract for Week 3; once confirmed, only the orchestrator's mock loader needs to change to plug in real data.

✔ Frontend — Design & Stack Finalized

— Neil (planning), Member 4 (build, starting)

- Full UI designed via Stitch across 3 screens (Landing, Login, Dashboard) with a locked dark forensics-terminal design system (color tokens, IBM Plex Mono + Inter typography)
- Nav wired for multi-page/click-through flow between screens; image content and interaction notes added in a refinement pass
- Final tech stack locked with exact dependency versions: React 19.2.7, Vite 8.1.4, TypeScript 7.0.2, Tailwind CSS 4.3.2, React Router 7.18.1, Zustand 5.0.14, TanStack Query 5.101.2, GSAP 3.15.0 (replacing Framer Motion per team decision), @xyflow/react 12.11.2
- GSAP plugin set finalized: ScrambleText, ScrollTrigger, DrawSVG, SplitText, useGSAP — mapped against 8 distinct animation moments across the app (hero decrypt effect, scroll reveals, hover states, page transitions, status pulse, IOC graph line-draw, behavioral summary text reveal)
- 6-day hand-coded build plan structured (no AI page-builders), scaffold → landing → animations → login (mock auth) → dashboard (mock data) → polish/transitions

✔ Dashboard Feature Set Expanded

— Member 4 (core build), Neil (AI Recommendations support)

Team input incorporated — added to Week 4 scope:

Feature	Tier	Notes
Process Tree	Core	Renders CAPE's native parent-child process data — no new backend work
Risk Score	Core	Already implemented in orchestrator skeleton (see above)
Evidence Timeline	Core	Chronological view of existing pipeline timestamps
Chain of Custody module	Core (differentiator)	Confirmed as its own visible dashboard tab, not buried backend logic
Geo-IP Map	Stretch	Strengthens India-angle narrative (domestic vs. foreign exfil visualization)
AI Recommendations	Stretch	Extension of existing narrative agent prompt, not a new agent
Malware Family tag	Stretch	Cheap version — surface matched YARA/family label, not full ML classification

✔ Project Documentation

— Neil

Full README.md written and delivered covering problem understanding, differentiators, architecture, complete tech stack, repo structure, setup instructions, team ownership, roadmap, and deployment model.

✔ Development Tooling

— Neil

opencode CLI (AI coding agent, terminal/VS-Code-integrated) installed and connected to a Google model for backend/infra assistance — explicitly scoped to backend/DevOps work only, frontend remains fully hand-coded per the team's earlier decision.

Week 2 — In Progress (tracked, not yet confirmed complete at time of writing)

Task	Owner	Status
Static analysis engine (hashing, YARA, Androguard/jadx, PE inspection)	Member 1	In progress
String/URL/IP extraction, end-to-end both platforms	Member 1	In progress
Ingestion gateway + isolation controller (Docker)	Member 2	In progress
KVM + CAPE install, Windows 10 golden snapshot on GCP	Member 2	In progress

Milestone Target — Thursday 16 July Meeting

Static pipeline demo on 1 APK + 1 PE sample; CAPE detonating live on the GCP sandbox. Status to be confirmed at the meeting and logged in Week 3 documentation.

